Hierarchical Bayesian Modeling of NFL First Down Success Probability

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Abstract

Predicting NFL first down success involves dynamic game situations and varying team abilities. This paper develops and evaluates a Bayesian Hierarchical Model (BHM) for estimating in-play first down probability (FDP) using 2023-2024 nflverse data. The BHM incorporates situational fixed effects (down, distance, field position, play type) and estimates team-specific offensive and defensive random effects hierarchically using NumPyro with NUTS sampling. Compared to a frequentist logistic regression baseline, both models show similar predictive accuracy (AUC $\approx$ 0.68 Bayesian vs. 0.67 Frequentist), but the BHM offers slightly better calibration (Brier Score 0.2213 vs. 0.2224). Posterior analysis quantifies the impact of situational factors and significant team heterogeneity. Scenario analyses demonstrate the models contextual responsiveness, notably capturing strategic factors in 4th and 1 goal-line situations. While predictive gains are marginal, the BHM provides enhanced interpretability, direct estimation of latent team abilities with uncertainty quantification, and a robust probabilistic framework valuable for sports analytics.

1 Introduction

Estimating the probability of success for discrete events within a dynamic sports context, such as achieving a first down in the National Football League (NFL), presents significant modeling challenges. The outcome depends heavily on the immediate game state (down, distance, field position) but is also modulated by latent factors like inherent team strengths, opponent capabilities, and strategic choices (e.g., pass or run). While traditional methods like standard logistic regression can capture the influence of observable predictors, they often struggle to adequately account for structured variability between teams or provide intuitive measures of uncertainty around predictions and parameter estimates. Bayesian Hierarchical Models (BHMs) offer a powerful and principled alternative (Gelman et al., 2013). By structuring parameters hierarchically (e.g., modeling individual team effects as draws from a common league-wide distribution), BHMs allow for partial pooling of information. This mechanism enables more stable estimation for entities with sparse data (shrinkage) while still permitting entity-specific deviations from the overall mean. Furthermore, the Bayesian paradigm inherently yields full posterior distributions for all parameters and predictions, naturally quantifying uncertainty. The code and analysis are available at: `https://colab.research.google.com/drive/17cOkeBYNhaNiT7_mb1b_3omADkZZNJYd?usp=sharing`.

This study applies modern Bayesian inference techniques to model in-play NFL first down success probability (FDP) using play-by-play data from the 2023-2024 NFL seasons. We develop a BHM incorporating situational fixed effects and team-specific random effects for both offensive and defensive prowess and compare its inferential richness and predictive performance against a standard frequentist logistic regression model. This work addresses the following research questions:

1. What are the key situational drivers of first down success probability and their magnitudes?
2. How much variability exists between NFL teams in their underlying offensive ability to convert first downs and their defensive ability to prevent them, after accounting for game situation?
3. How do the predictive performance and, more importantly, the interpretability and uncertainty quantification of the BHM compare to a standard frequentist approach for this problem?

4. Can this BHM provide a flexible, probabilistic framework for FDP estimation applicable to nuanced sports analytics?

2 Data Acquisition and Pre-processing

The analysis utilizes publicly available play-by-play data for the 2023 and 2024 NFL regular seasons, sourced from the nflverse data repository (Tan Ho et al., 2022). This dataset provides granular information for each play.

Initial data processing involved selecting relevant variables encompassing game situation (`down`, `ydstogo`, field position via `yardline_100`, `play_type`) and the plays outcome relative to achieving a first down (`series_result`). Data cleaning and filtering steps included:

- Retaining only regular season (`season_type == REG`) games.
- Focusing on standard offensive plays: only plays explicitly classified as run or pass were included. Special teams plays, non-standard plays, and plays where the first down outcome was determined by an accepted penalty were excluded.
- Removing records with missing values in essential variables (`down`, `ydstogo`, `yardline_100`, `posteam`, `defteam`, `series_result`) to ensure data integrity.

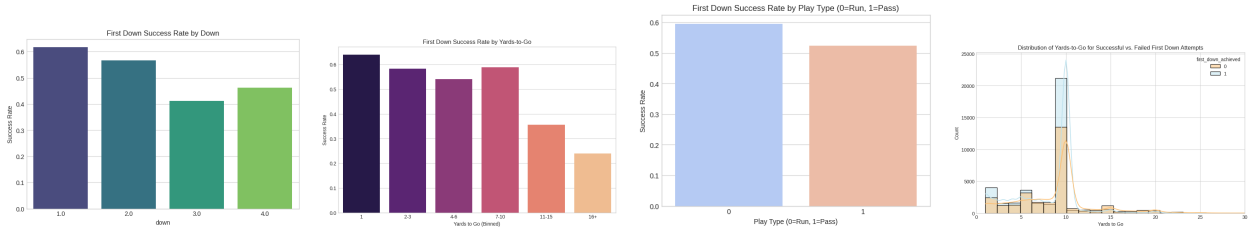
Feature engineering was performed to create the necessary model variables:

- **Target Variable:** A binary outcome variable, `first_down_achieved`, was created, coded as 1 if the `series_result` indicated a First down, and 0 otherwise.
- **Predictor Variable:** A binary indicator, `is_pass`, was created, coded as 1 if the `play_type` was pass, and 0 if it was run.

To prevent data leakage and ensure realistic evaluation, the data was split into training (80%) and testing (20%) sets based on unique `game_id`. This guarantees that all plays from a single game reside entirely within one set. For the Bayesian model, continuous predictors (`down`, `ydstogo`, `yardline_100`) were standardized (centered and scaled to unit variance) using scikit-learns `StandardScaler`.

3 Data Exploration

Exploratory Data Analysis (EDA) was conducted on the cleaned training dataset to understand marginal relationships prior to formal modeling. Summary statistics revealed an overall first down conversion rate of approximately 55.6%. Visualizations (Figure 1) provided further insights. Success rate clearly decreases from 1st down (≈62%) through 3rd down (≈41%), with a slight uptick on 4th down (≈46%) (Fig 1a). A strong negative relationship exists with yards-to-go (Fig 1b), with success rates exceeding 65% for 1 yard needed but falling sharply with distance. Aggregate data shows runs have a slightly higher success rate (≈59%) than passes (≈52%) (Fig 1c), though this ignores context. The distribution of yards-to-go (Fig 1d) shows successful attempts peak at short yardage and around 10 yards, while failures are more common at 10+ yards. These findings confirm the relevance of the selected predictors.



(a) Down (b) YdsToGo (c) Play Type (d) YdsToGo Dist

Figure 1: Exploratory Data Analysis Visualizations (Training Data).

4 Prior Literature

The increasing availability of granular play-by-play NFL data, accessible through resources like nflverse (Tan Ho et al., 2022), enables sophisticated modeling of in-game events. While generalized linear models (GLMs) are common for predicting outcomes like play success, they often struggle to fully capture team-specific variations in ability or provide comprehensive uncertainty estimates essential for decision-making.

Bayesian Hierarchical Models (BHMs) provide a powerful framework to address these limitations (Gelman et al., 2013). By structuring parameters in a hierarchy—for instance, modeling individual team strengths as varying around a league-wide average—BHMs naturally incorporate team-level variability. This hierarchical structure facilitates "partial pooling" of information across teams, leading to more stable and sensible estimates, especially for teams with less data, by shrinking estimates towards the overall mean (Baio and Blangiardo, 2010). A key strength of the Bayesian framework is its output: full posterior distributions for every parameter and prediction. This provides a direct and intuitive way to quantify uncertainty. BHMs have demonstrated success in various sports contexts, such as modeling soccer goal rates, precisely because they can estimate latent abilities while accounting for context (Baio and Blangiardo, 2010).

While the specific modeling of first down success is less common in literature compared to broader metrics like Expected Points Added (EPA) or Win Probability (WP), the fundamental challenge of separating situational effects from intrinsic team ability persists. A hierarchical Bayesian approach is particularly well-suited to estimating these underlying offensive and defensive team strengths related to first down conversions, offering a more nuanced view than models relying solely on fixed effects or simple team adjustments. This study applies a modern BHM framework, leveraging efficient Hamiltonian Monte Carlo sampling (NumPyro), to the specific problem of NFL FDP. We focus on explicitly estimating hierarchical team effects and comparing the predictive performance and interpretive advantages, especially concerning uncertainty and latent ability estimation, against a standard frequentist baseline.

5 Proposed Method: Model Specification

Two modeling approaches were employed to estimate the probability of achieving a first down (`first_down_achieved` = 1), allowing for comparison and addressing the research questions regarding driver identification, team effects, and model utility.

5.1 Frequentist Model (Logistic Regression)

A standard binomial logistic regression served as a baseline model. The probability of success, p_i, for play i was modeled using the logit link function dependent on situational predictors:

$$\text{logit}(p_i) = \beta_0 + \beta_1 \text{down}_i + \beta_2 \text{ydstogo}_i + \beta_3 \text{is_pass}_i + \beta_4 \text{yardline_100}_i$$

This model was fitted using maximum likelihood estimation via the `statsmodels` library in Python, utilizing the training dataset. It assumes predictor effects are constant across all teams and does not incorporate team-specific parameters, providing a benchmark for the BHMs handling of heterogeneity.

5.2 Bayesian Hierarchical Model

A Bayesian hierarchical logistic regression was specified to account for both situational factors and team-specific variability, directly addressing research questions about team effects and uncertainty. The likelihood for the binary outcome y_i for play i involving possessing team j and defending team k is assumed to be Bernoulli:

$$y_i \sim \text{Ber}(p_i)$$

The probability of success p_i is linked to a linear predictor via the logit function:

$$\text{logit}(p_i) = \underbrace{\beta_0 + \boldsymbol{\beta}^T \mathbf{X}_i}_{\text{Fixed Effects}} + \underbrace{\alpha_{\text{off},j} - \alpha_{\text{def},k}}_{\text{Random Effects}}$$

where:

- $\mathbf{X}_i$ is the vector of scaled situational predictors for play i (`down`, `ydstogo`, `yardline_100`, `is_pass`), capturing the game context (RQ1).
- β_0 is the global intercept.
- $\boldsymbol{\beta}$ is the vector of coefficients for the fixed effects.
- $\alpha_{\text{off},j}$ is the random effect representing the offensive strength of team j (RQ2).
- $\alpha_{\text{def},k}$ is the random effect representing the defensive strength of team k (RQ2). A higher value implies a weaker defense.

The rationale for the hierarchical structure is to explicitly model team heterogeneity while leveraging partial pooling for robust estimation. Team-specific random effects are modeled as draws from common distributions:

$$\alpha_{\text{off},j} \sim \mathcal{N}(\mu_{\text{off}}, \sigma^2_{\text{off}})$$
$$\alpha_{\text{def},k} \sim \mathcal{N}(\mu_{\text{def}}, \sigma^2_{\text{def}})$$

For improved sampling efficiency (a computational detail), a non-centered parameterization was implemented:

$$\alpha_{\text{off},j} = \mu_{\text{off}} + \sigma_{\text{off}} \cdot \alpha^{\text{raw}}_{\text{off},j} \quad \text{where} \quad \alpha^{\text{raw}}_{\text{off},j} \sim \mathcal{N}(0,1)$$
$$\alpha_{\text{def},k} = \mu_{\text{def}} + \sigma_{\text{def}} \cdot \alpha^{\text{raw}}_{\text{def},k} \quad \text{where} \quad \alpha^{\text{raw}}_{\text{def},k} \sim \mathcal{N}(0,1)$$

Prior Specification: Weakly informative priors were assigned to allow the data to drive the posterior while providing mild regularization:

- Intercept: $\beta_0 \sim \mathcal{N}(0, 2^2)$
- Fixed Effects Coefficients: $\beta_p \sim \mathcal{N}(0, 2^2)$ for each predictor p.
- Random Effect Means: $\mu_{\text{off}} \sim \mathcal{N}(0, 1^2)$, $\mu_{\text{def}} \sim \mathcal{N}(0, 1^2)$
- Random Effect Standard Deviations: $\sigma_{\text{off}} \sim \text{HalfNormal}(1)$, $\sigma_{\text{def}} \sim \text{HalfNormal}(1)$ (ensuring non-negativity for SDs).
- Raw Random Effects: $\alpha^{\text{raw}}_{\text{off},j} \sim \mathcal{N}(0,1)$, $\alpha^{\text{raw}}_{\text{def},k} \sim \mathcal{N}(0,1)$

6 Proposed Method: Computational Solution

6.1 Hamiltonian Monte Carlo (HMC) & No-U-Turn Sampler (NUTS)

Posterior distributions for the BHM parameters were estimated using Markov Chain Monte Carlo (MCMC) via the NumPyro library. We employed the No-U-Turn Sampler (NUTS), an adaptive variant of Hamiltonian Monte Carlo (HMC). HMC is a gradient-based MCMC algorithm that leverages physics-inspired dynamics to explore the parameter space efficiently. It introduces auxiliary momentum variables and simulates the Hamiltonian dynamics of a conceptual particle moving on the potential energy surface defined by the negative log-posterior density. This simulation, typically performed using numerical integration like the leapfrog method, allows HMC to propose distant states with high acceptance probability, mitigating the random-walk behavior of simpler samplers and improving efficiency in high-dimensional or correlated posteriors. Proposals are accepted or rejected based on the conservation of the simulated Hamiltonian energy. NUTS enhances HMC by automatically and adaptively tuning the number of leapfrog steps (path length) for each proposal. It does this by simulating forward and backward in time until the trajectory begins to make a "U-turn," indicating that further simulation might be inefficient. This avoids the need for manual tuning of the path length parameter and prevents inefficient backtracking, making NUTS a robust and effective sampler for complex, differentiable models.

6.2 MCMC Settings & Implementation

The NUTS sampler was executed using NumPyro. We ran 2 independent MCMC chains. Each chain involved 1000 warmup (burn-in) iterations (discarded) followed by 1500 sampling iterations retained for analysis, resulting in a total MCMC sample size of 3000 from the joint posterior distribution.

6.3 Sampling Diagnostics

MCMC convergence and sample quality were assessed using standard diagnostics:

- **Potential Scale Reduction Factor ($\hat{R}$):** Values were ≈ 1.00 for all key parameters (fixed effects, hyperparameters), providing strong evidence that both chains converged to the same target distribution.
- **Effective Sample Size (ESS):** Bulk and tail ESS values were sufficiently high for all primary parameters (e.g., `ess_bulk` > 750 for σ parameters, > 2400 for β and μ parameters), ensuring reliable posterior summary estimates.
- **Trace Plots:** Visual inspection of trace plots (Figure 2) for key parameters showed stationary, well-mixed chains appearing as horizontal "fuzzy caterpillar" bands with good overlap. Autocorrelation plots (not shown) demonstrated rapidly decaying autocorrelation, confirming NUTS efficiency.

Collectively, these diagnostics confirm that the MCMC simulation successfully converged and generated high-quality samples suitable for posterior inference.

7 Data analysis and results interpretation

7.1 Frequentist Model Results

The baseline logistic regression (fitted on training data) showed all predictors were statistically significant (p ¡ 0.001). Increased `down` ($\beta \approx -0.54$) and `distance` ($\beta \approx -0.13$) decreased success odds, as did passing plays (`is_pass`, $\beta \approx -0.08$). Counter-intuitively, odds increased further from the opponents endzone (`yardline_100`, $\beta \approx 0.02$).

7.2 Bayesian Model: Fixed Effects & Hyperparameters

Bayesian posterior summaries (visualized in Figure 2) largely mirrored frequentist point estimates for fixed effects. Key posterior means and 94% Highest Density Intervals (HDIs) were:

- Intercept (β_0): Mean 0.14, HDI [-2.10, 2.21].
- Down (β_1): Mean -0.45, HDI [-0.47, -0.43].
- Ydstogo (β_2): Mean -0.52, HDI [-0.54, -0.50].
- Yardline (β_4): Mean 0.52, HDI [0.50, 0.53].
- Is_Pass (β_3): Mean -0.07, HDI [-0.11, -0.04].

All fixed effect HDIs exclude zero, confirming credible effects ($\hat{R}$=1.00, ESS > 2400 for all). The positive `yardline_100` effect persists. For hyperparameters (capturing league-wide tendencies):

- Mean Offense Effect (μ_{off}): Mean 0.07, HDI [-1.62, 1.72].
- Offense Std Dev (σ_{off}): Mean 0.14, HDI [0.10, 0.17].
- Mean Defense Effect (μ_{def}): Mean -0.06, HDI [-1.74, 1.65].
- Defense Std Dev (σ_{def}): Mean 0.10, HDI [0.07, 0.13].

Team effect means are centered near zero (HDIs span zero), while standard deviations are credibly positive, quantifying moderate team heterogeneity (larger for offense; $\hat{R}$=1.00, ESS > 750 for σ, > 2400 for μ).

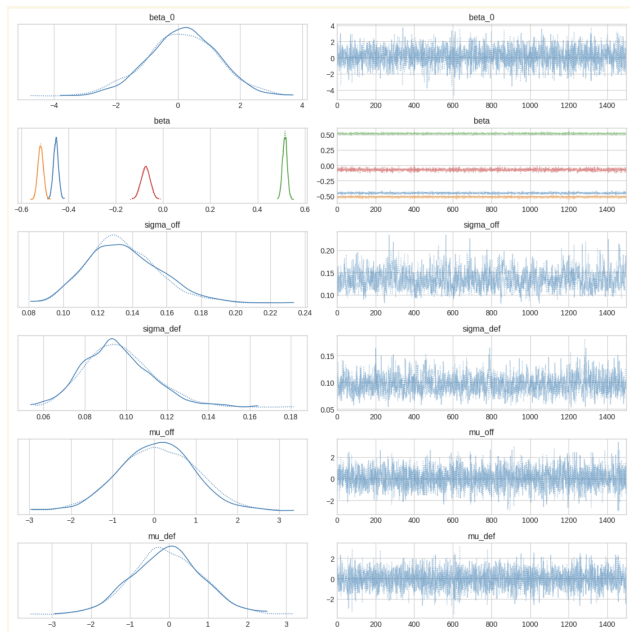


Figure 2: Trace plots (right) and posterior densities (left) for key Bayesian parameters.

7.3 Bayesian Model: Team Effects

Estimated team-specific offensive (α_{off}) and defensive (α_{def}) effects, representing deviations from the league average after accounting for situation, are shown in Figures 3 and 4.

- **Offense:** Teams like KC, PHI, DET, BUF exhibit posterior distributions shifted towards higher positive values, indicating stronger first-down conversion ability relative to average. Conversely, NE, CLE, NYJ show distributions shifted lower.
- **Defense:** Teams like CLE, NYJ, NO show distributions shifted towards lower values (indicating better-than-average prevention). Teams like KC, BUF, PHI have distributions shifted higher (weaker prevention). The width of each violin reflects the posterior uncertainty in that teams estimated effect.

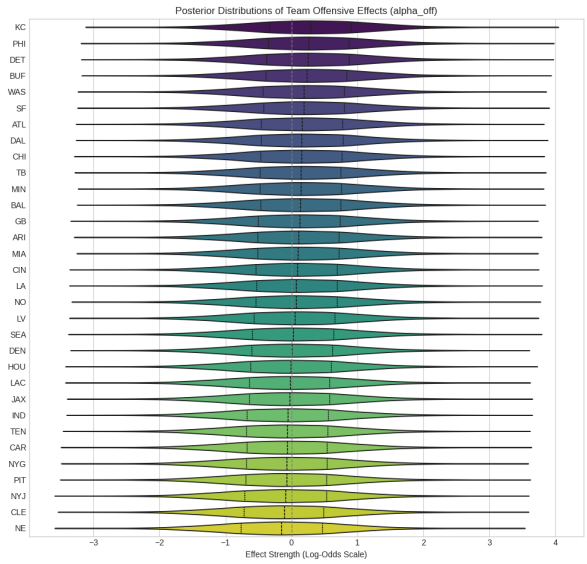


Figure 3: Posterior Distributions of Team Offensive Effects (α_{off}).

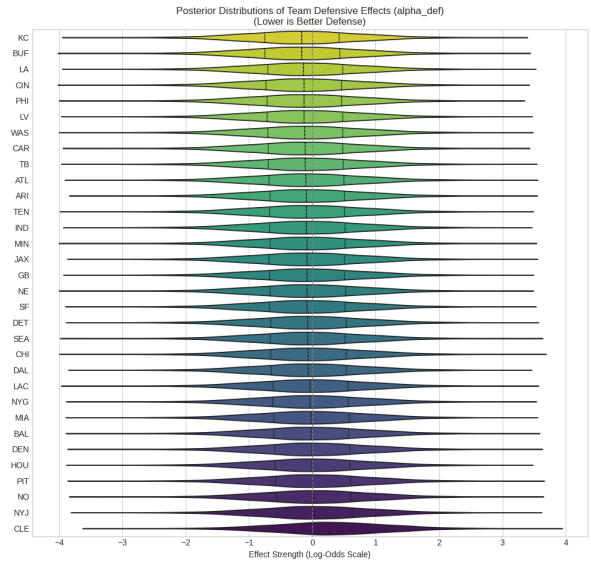


Figure 4: Posterior Distributions of Team Defensive Effects (α_{def}). Lower is better.

7.4 Model Evaluation

Performance on the held-out test set (Table 1) shows both models achieve similar accuracy (≈0.65) and AUC (≈0.67-0.68), indicating moderate discriminative power (Figures 5 and 6). The BHM shows slightly better calibration (lower Brier Score: 0.2213 vs. 0.2224; lower Log Loss: 0.6331 vs. 0.6356) and marginally better classification trade-offs (more TPs, fewer FNs).

Table 1: Model Evaluation Metrics on Test Set

Metric	Frequentist	Bayesian
Accuracy	0.6483	0.6499
AUC	0.6738	0.6778
Brier Score	0.2224	0.2213
Log Loss	0.6356	0.6331
True Negatives (TN)	2998	2955
False Positives (FP)	2913	2956
False Negatives (FN)	1815	1751
True Positives (TP)	5718	5782

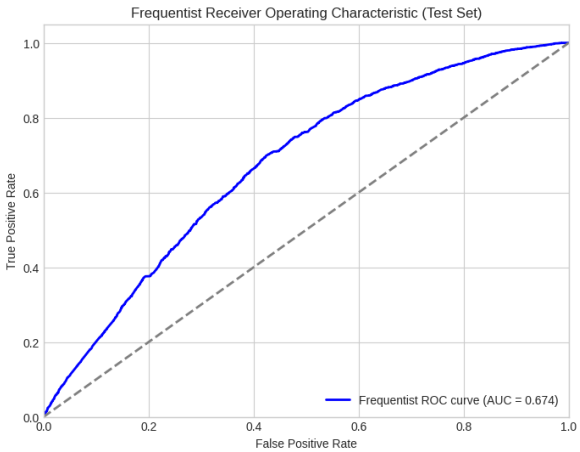


Figure 5: Frequentist ROC Curve (Test Set).

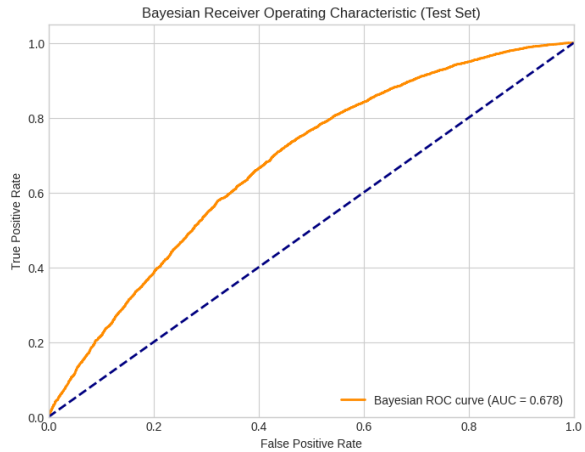


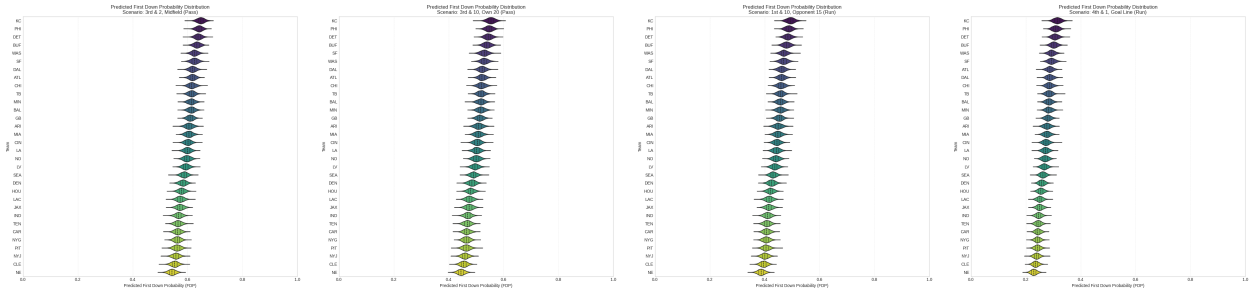
Figure 6: Bayesian ROC Curve (Test Set).

7.5 Scenario Analysis

Predicted FDP distributions for specific scenarios against a median defense (Figure 7) highlight contextual effects and the BHMs ability to generate probabilistic forecasts:

- **3rd & 2, Midfield (Pass):** Highest FDPs (medians ≈0.55-0.70); top offenses excel.
- **3rd & 10, Own 20 (Pass):** Lower FDPs (medians ≈0.45-0.58).
- **1st & 10, Opponent 15 (Run):** FDPs generally lower (medians ≈0.40-0.55).
- **4th & 1, Goal Line (Run):** Lowest predicted FDPs (medians ≈0.18-0.30), likely reflecting strategic risk and observed low success rates in the data for these high-leverage plays, not just distance.

The distributions show the range of likely outcomes for each team in each situation.



(a) 3rd&2, Mid (P) (b) 3rd&10, Own20 (P) (c) 1st&10, Opp15 (R) (d) 4th&1, Goal (R)

Figure 7: Predicted FDP Distributions for Specific Scenarios (vs. Median Defense). Letters indicate Play Type.

8 Discussion

This analysis confirms that situational factors like down and distance strongly drive NFL first down success. The BHM effectively captured this while quantifying significant team-level heterogeneity in offensive conversion ($\sigma_{\text{off}} \approx 0.14$) and defensive prevention ($\sigma_{\text{def}} \approx 0.10$). These estimates offer interpretable team strength rankings specific to first downs, complete with uncertainty bounds (Figures 3, 4), addressing key research questions.

Comparing the BHM to the frequentist baseline highlights its advantage in interpretability. While overall AUC was similar (≈0.68 vs. ≈0.67), the BHM directly estimates latent team abilities ($\alpha_{\text{off}}, \alpha_{\text{def}}$) with uncertainty. This enables nuanced assessments of team performance relative to the league average within

context. The scenario analysis (Figure 7) further demonstrates this by providing probabilistic forecasts, offering richer insights than point estimates. Slightly better calibration metrics (Brier, Log Loss) also favor the BHM. Thus, while not dramatically improving aggregate accuracy, the BHM provides more actionable and interpretable information for understanding performance and making probabilistic assessments.

However, some model limitations warrant attention. The persistent positive coefficient for `yardline_100` (suggesting higher success further from the opponents endzone) is counter-intuitive. This likely indicates omitted variables (like score differential or time) or perhaps reflects increased defensive intensity (prevent defense) closer to their own goal line, which isn't fully captured. Similarly, the very low FDP predicted in 4th and 1 goal-line situations (Fig 7d) reflects observed low success rates influenced by defensive stacking and play-call predictability, not just distance. The model also assumes play independence and constant team effects over the analyzed seasons, and using only two recent seasons limits data volume, potentially affecting precision for some teams. Results are conditional on the chosen priors.

9 Conclusion

This study successfully applied a Bayesian hierarchical model to the complex problem of predicting NFL first down success, demonstrating its ability to integrate situational context while estimating latent team offensive and defensive strengths with associated uncertainty. Compared to a standard frequentist logistic regression, the BHM offered similar predictive accuracy but provided superior interpretability, direct quantification of latent team abilities, full probabilistic predictions for scenario analysis, and slightly improved calibration. These advantages highlight the utility of BHMs for gaining nuanced, uncertainty-aware insights into complex sports data, particularly when the goal is to understand underlying abilities and contextual dependencies beyond simple point prediction.

References

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